
Saad Ahmad Khan

IISER Tirupati 517619 • saadahmad@students.iisertirupati.ac.in • +91 8600626736

Research Interests:

Structural Biology • Protein Biochemistry • Membrane Proteins • Cryo-EM • Biomolecular Condensates • Protein Assembly • Molecular Biophysics

Experience

**National Centre for Cell Science (NCCS)
MS - THESIS**

Pune, MH
July 2024 - April 2025

- Optimised E. coli expression and multi-step purification (affinity, reverse-phase HPLC, size-exclusion chromatography) protocol of nucleoporin IDP constructs, achieving high-purity protein yield via Lyophilization.
- Demonstrated NaCl-induced phase separation of IDP constructs via centrifugation assays.
- Applied ImageJ/Fiji for quantitative image analysis and IUPred for disorder prediction to correlate sequence features with observed phase behavior
- Engineered site-directed variants of IDR sequences to add cystine for maleimide labeling of protein and DIC/fluorescence microscopy for visualization of PEG induced protein condensate formation.
- Executed experiment independently, encompassing experimental design, data analysis and thesis writing.

**Indian Institute of Science Education and Research (IISER)
Semester Project**

Tirupati, AP
August 2023 - November 2023

- Cloning of genes related to memory in heat stress primed plant

Bioinformatics Lab

- Computational Biology
- January 2023 - April 2023

Technical University of Denmark (DTU) - Online (Coursera)

- Computational course on Whole genome sequencing of bacterial genomes - tools and applications.
- December 2023
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Events and Workshops

2025 How to Grow Almost Anything (MIT Media Lab) - Online (Global Listener)

- Synthetic biology bootcamp, Synthetic Minimal Cells, DNA Nanostructures, Next Generation Synthesis etc March - May 2025

India| EMBO Lecture Course (NCCS Pune)

- RNA-protein complexes: from molecular assembly to physiological functions and disease Pune, MH
24-28 February 2025

Biophysics Paschim -2024 (IISER Pune)

- Biophysics Conference Pune, MH
24 August 2024

SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY

- Workshop on Advanced Industrial Biotechnology Techniques Tirupati, AP
1-3 November 2023

Skills

Protein Biochemistry:

Protein expression, Affinity chromatography, Size exclusion chromatography, Reverse phase HPLC, Protein purification, Protein quantification

Molecular Biology:

Cloning, PCR, Site-directed mutagenesis

Biophysical Characterization:

Protein phase separation, Fluorescence microscopy, DIC microscopy, ImageJ, Fluorescence spectroscopy, Absorbance spectroscopy

Computational:

R, Python

Reference

Dr. Radha Chauhan
(MS Thesis Supervisor)
Lab of Macromolecular Assembly Dynamics
NCCS, Pune
Email: radha.chauhan@nccs.res.in

Dr. Ravi Kumar Pujala
(Faculty Advisor)
Associate Professor, Physics
IISER Tirupati
Email: pujalaravikumar@labs.iisertirupati.ac.in

Dr. Annapurna Devi Allu
(Semester Project Supervisor)
Associate Dean, International Relations
IISER Tirupati
Email: anu.allu@labs.iisertirupati.ac.in